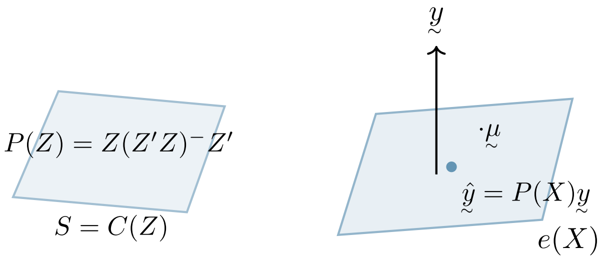
W03 L02

Q. For which $\underset{\sim}{\beta}$, $P(X)\underset{\sim}{y} = X\underset{\sim}{\beta}$?

$$\Rightarrow \; X(X'X)^{-}X'\underset{\sim}{y} = X\underset{\sim}{\beta} \qquad (*)$$



Case 1: $(X'X)^{-1}$ exists

$\Leftrightarrow$ X has full column rank, $\rho(X) = p+1$

$\Leftrightarrow$ There exists a matrix Z such that $ZX = I_{p+1}$. (Left inverse)

So, pre-multiply $(*)$ by Z,

$$\underbrace{ZX}_{I_{p+1}}(X'X)^{-1}X'\underset{\sim}{y} = \underbrace{ZX}_{I_{p+1}}\underset{\sim}{\beta} = \hat{\beta}_{\text{OLS}}.$$

Case II: $X'X$ is singular

Recall the problem of finding solution. of $A\underset{\sim}{x} = \underset{\sim}{b}$ where A is not invertible.

$X\underset{\sim}{\beta} = C$ where $C = X(X'X)^{-}X'\underset{\sim}{y}, \quad n > p+1$.

Here unique solution does not exist. One possible soln. is

$$\hat{\underset{\sim}{\beta}} = (X'X)^{-}X'\underset{\sim}{y}.$$

Residual: $\hat{\underset{\sim}{\varepsilon}} = \underset{\sim}{y} - X\hat{\underset{\sim}{\beta}}$.

$\hat{\underset{\sim}{\beta}}_{\text{OLS}} = (X'X)^{-}X'\underset{\sim}{y}$: Choice of $\underset{\sim}{\beta}$ which minimizes $\underset{\sim}{\varepsilon}^T\underset{\sim}{\varepsilon}$.

The minimized $\underset{\sim}{e}$ is

$$\underset{\sim}{y} - X\hat{\underset{\sim}{\beta}}_{\text{OLS}} = \big(I - P(X)\big)\underset{\sim}{y} = \underset{\sim}{e} \quad \text{(residual)}.$$

Minimized $\underset{\sim}{\varepsilon}^T\underset{\sim}{\varepsilon}$ is

$$\|\underset{\sim}{y} - X\hat{\underset{\sim}{\beta}}_{\text{OLS}}\|^2 = \underset{\sim}{y}^T\big(I - P(X)\big)\underset{\sim}{y} \quad \text{(RSS residual Sum of Squares)}.$$

Remark: When $(X'X)$ is singular, then the soln. to least squares eqn. (i.e., $\underset{\sim}{\beta}$ in $(*)$) is not unique.
However, as $P(X)$ is unique in this case as well, $\hat{\underset{\sim}{y}} = P(X)\underset{\sim}{y}$, $\underset{\sim}{e} = \big(I - P(X)\big)\underset{\sim}{y}$ and RSS are unique.

Estimation of σ^2

Result

$\dfrac{\text{RSS}}{n-r}$, where $\rho(X) = r$, is an unbiased estimator of σ^2.

Proof:

$$E\Big(\frac{\text{RSS}}{n-r}\Big) = E\Big(\frac{\underset{\sim}{y}^T\big(I - P(X)\big)\underset{\sim}{y}}{n-r}\Big)$$
$$= \frac{1}{n-r}\Big\{(X\underset{\sim}{\beta})^T\big(I - P(X)\big)(X\underset{\sim}{\beta}) + \text{tr}\big(\sigma^2 I\big(I - P(X)\big)\big)\Big\}$$

$\big[z\text{As } X\underset{\sim}{\beta} \in e(X),\ \big(I - P(X)\big)X\underset{\sim}{\beta} = 0.\big]$

Recall: PS 1:

$E(\underset{\sim}{x}^T A\underset{\sim}{x}) = \underset{\sim}{\mu}^T A\underset{\sim}{\mu} + \text{tr}(A\Sigma)$ when $E(\underset{\sim}{x}) = \underset{\sim}{\mu}$, $\text{var}(\underset{\sim}{x}) = \Sigma$.

$$= \frac{1}{n-r}\Big\{0 + \sigma^2\big(\text{tr}(I_n) - \text{tr}(P(X))\big)\Big\}$$
$$= \sigma^2,$$

since $\text{tr}(P(X)) = \rho(P(X)) = \rho(X) = r$.

We write $\hat{\sigma}^2 = \dfrac{\text{RSS}}{n-r}$.

Some Properties of $\hat{\underset{\sim}{\beta}}_{\text{OLS}}$

1) $E_\beta(\hat{\underset{\sim}{\beta}}_{\text{OLS}}) = \underset{\sim}{\beta}$. (Prove this when $(X'X)^{-1}$ exists)

2) $\text{Var}_\beta(\hat{\underset{\sim}{\beta}}_{\text{OLS}}) = \sigma^2(X'X)^{-1}$ (" ")

$$\text{Var}(\hat{\underset{\sim}{\beta}}_{\text{OLS}}) = \text{Var}\big((X'X)^{-1}X'\underset{\sim}{y}\big) = (X'X)^{-1}X'\ \text{var}(\underset{\sim}{y})\, X(X'X)^{-1}$$
$$= \sigma^2(X'X)^{-1}.$$

3) Let $\underset{\sim}{c}^T\underset{\sim}{\mu} = g(\underset{\sim}{\mu})$ is the parameter of interest, where $\underset{\sim}{\mu} = X\underset{\sim}{\beta}$, and $\hat{\underset{\sim}{\mu}}_{\text{OLS}} = X\hat{\underset{\sim}{\beta}}_{\text{OLS}}$. Then $\underset{\sim}{c}^T\hat{\underset{\sim}{\mu}}_{\text{OLS}}$ is the BLUE (best linear unbiased estimator) of $g(\underset{\sim}{\mu})$ (even if $(X'X)$ is singular).

Proof:

$$\underset{\sim}{c}^T \hat{\underset{\sim}{\mu}}_{\text{OLS}} = \underset{\sim}{c}^T X(X'X)^- X' \underset{\sim}{y} \quad (\text{linear in } \underset{\sim}{y}).$$

$$E(\underset{\sim}{c}^T \hat{\underset{\sim}{\mu}}_{\text{OLS}}) = \underset{\sim}{c}^T \underbrace{X(X'X)^- X'}_{P(X)} \underbrace{X\underset{\sim}{\beta}}_{\mu \in C(X)} = \underset{\sim}{c}^T \underset{\sim}{\mu} \quad (\text{unbiased}).$$

Let $\underset{\sim}{a}^T \underset{\sim}{y}$ be another LUE of $\underset{\sim}{c}^T \underset{\sim}{\mu}$.

As $\underset{\sim}{a}^T \underset{\sim}{y}$ is unbiased, $E(\underset{\sim}{a}^T \underset{\sim}{y}) = \underset{\sim}{a}^T \underset{\sim}{\mu} = \underset{\sim}{c}^T \underset{\sim}{\mu}$.

$$\begin{aligned}
&\Rightarrow (\underset{\sim}{a} - \underset{\sim}{c})^T \underset{\sim}{\mu} = (\underset{\sim}{a} - \underset{\sim}{c})^T X \underset{\sim}{\beta} = 0 \\
&\Rightarrow (\underset{\sim}{a} - \underset{\sim}{c})^T X = \underset{\sim}{0}^T \quad \big(\text{as } \underset{\sim}{\beta} \in \mathbb{R}^{p+1} \text{ is any arbitrary vector}\big) \\
&\Rightarrow (\underset{\sim}{a} - \underset{\sim}{c})^T P(X) = \underset{\sim}{0}^T \\
&\Rightarrow P(X)(\underset{\sim}{a} - \underset{\sim}{c}) = \underset{\sim}{0} \qquad (*).
\end{aligned}$$

$$\begin{aligned}
\text{Var}(\underset{\sim}{a}^T \underset{\sim}{y}) - \text{Var}(\underset{\sim}{c}^T \hat{\underset{\sim}{\mu}}_{\text{OLS}}) &= (\underset{\sim}{a}^T \underset{\sim}{a})\sigma^2 - \underset{\sim}{c}^T \text{Var}\big(P(X)\underset{\sim}{y}\big)\underset{\sim}{c} \\
&= \sigma^2 \left[\underset{\sim}{a}^T \underset{\sim}{a} - \underset{\sim}{c}^T P(X) \underset{\sim}{c}\right] \quad \big((P(X)\underset{\sim}{c})^T (P(X)\underset{\sim}{c} = \underset{\sim}{c}^T P(X)\underset{\sim}{c}\big) \\
&= \sigma^2 \left[\underset{\sim}{a}^T \underset{\sim}{a} - \underset{\sim}{a}^T P(X) \underset{\sim}{a}\right] \quad \big(P(X)\underset{\sim}{c} = (P(X)\underset{\sim}{a} \text{ by } (*)\big) \\
&= \sigma^2 \underset{\sim}{a}^T \big(I - P(X)\big) \underset{\sim}{a} = \sigma^2 \big\|(I - P(X))\underset{\sim}{a}\big\|^2 \\
&\geq 0.
\end{aligned}$$

Why? Equality holds, $\underset{\sim}{a} = P(X)\underset{\sim}{a} \overset{(*)}{=} P(X)\underset{\sim}{c}$.

Then $\underset{\sim}{a}^T \underset{\sim}{y} = \underset{\sim}{c}^T P(X) \underset{\sim}{y} = \underset{\sim}{c}^T \hat{\underset{\sim}{\mu}}_{\text{OLS}}$.

Distribution Theory

$$\underset{\sim}{\varepsilon} \sim N_n(\underset{\sim}{0}, \sigma^2 I) \;\Leftrightarrow\; \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

Result

Then $\underset{\sim}{Y} \sim N(X\underset{\sim}{\beta}, \sigma^2 I)$, $\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$.

i. $\hat{\underset{\sim}{\beta}}_{\text{OLS}} \sim N_{p+1}\big(\underset{\sim}{\beta}, \sigma^2 (X'X)^{-1}\big) \quad ((X'X)^{-1} \text{ exists}).$

ii. $\dfrac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}$.

$$\left[\frac{\text{RSS}}{\sigma^2} = \frac{\underset{\sim}{\varepsilon}^T\underset{\sim}{\varepsilon}}{\sigma^2} = \frac{1}{\sigma^2}\underset{\sim}{y}^T\big(I - P(X)\big)\underset{\sim}{y}\right]$$

$$\left[\underset{\sim}{Z} = \frac{\underset{\sim}{Y} - X\underset{\sim}{\beta}}{\sigma} \sim N(\underset{\sim}{0}, I)\right]$$

$$\begin{aligned}
&= \frac{1}{\sigma^2}\big(\underset{\sim}{Y} - X\underset{\sim}{\beta} + X\underset{\sim}{\beta}\big)^T\big(I - P(X)\big)\big(\underset{\sim}{Y} - X\underset{\sim}{\beta} + X\underset{\sim}{\beta}\big) \\
&= \frac{\underset{\sim}{\varepsilon}^T\big(I - P(X)\big)\underset{\sim}{\varepsilon}}{\sigma^2} + 0 \qquad \big(\big(I - P(X)\big)X\underset{\sim}{\beta} = 0\big) \\
&= \underset{\sim}{Z}^T\big(I - P(X)\big)\underset{\sim}{Z}.
\end{aligned}$$

$$\begin{aligned}
\underset{\sim}{Z} &\sim N(\underset{\sim}{0}, I), \\
\underset{\sim}{Z}^T A \underset{\sim}{Z} &\sim \chi^2_r, \qquad \text{if } A \text{ is symmetric, idempotent,} \\
\rho(A) &= r.
\end{aligned}$$

$$\big(I - P(X)\big)X\underset{\sim}{\beta} = 0.$$

$$X\underset{\sim}{\beta} = \mu \in C(X)$$

Symmetric, idempotent, $\rho\big(I - P(X)\big) = n - \rho(X) = n - (p+1)$.

HW: What happens if $\rho(X) = r < p + 1$?
Show that, in this case $\frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-r}$.

Maximum Likelihood Estimation

Log-likelihood,

$$\ell(\underset{\sim}{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(\underset{\sim}{Y} - X\underset{\sim}{\beta})^T(\underset{\sim}{Y} - X\underset{\sim}{\beta}).$$

Maximizing the likelihood w.r.t. $\underset{\sim}{\beta}$, is same as minimizing the negative of last part, which is $\|\underset{\sim}{Y} - X\underset{\sim}{\beta}\|^2$, $\sigma^2 > 0$.

Thus,

$$\ell(\underset{\sim}{\beta}, \sigma^2) \;\leq\; \ell(\hat{\underset{\sim}{\beta}}_{\text{OLS}}, \sigma^2).$$

Now, we can simply minimize $\ell(\hat{\underset{\sim}{\beta}}_{\text{OLS}}, \sigma^2)$ w.r.t. σ^2.

$$\ell(\underset{\sim}{\hat{\beta}}_{\text{OLS}}, \sigma^2) = h(\sigma^2) = C - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(\text{RSS}).$$

$$\frac{\partial \ell(\underset{\sim}{\hat{\beta}}_{\text{OLS}}, \sigma^2)}{\partial \sigma^2} = 0 \;\Rightarrow\; \frac{\text{RSS}}{n} = \sigma^2.$$

$$\begin{aligned}
\left.\frac{\partial^2 \ell(\underset{\sim}{\hat{\beta}}_{\text{OLS}}, \sigma^2)}{\partial (\sigma^2)^2}\right|_{\sigma^2=\frac{\text{RSS}}{n}} &= \frac{\partial}{\partial \sigma^2}\Big(-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\text{RSS}\Big)\Big|_{\sigma^2=\frac{\text{RSS}}{n}} \\
&= \Big(\frac{n}{2\sigma^4} - \frac{\text{RSS}}{\sigma^6}\Big)\Big|_{\sigma^2=\frac{\text{RSS}}{n}} \\
&= \Big(\frac{n^3}{2\,\text{RSS}^2} - \frac{n^3}{\text{RSS}^2}\Big) = -\frac{n^3}{2\,\text{RSS}^2} < 0.
\end{aligned}$$